Reg. No. | | | | | | | | | | | [14EC64]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE-2019

DIGITAL VLSI DESIGN

Time: 3 Hours **Maximum Marks:70**

Instructions: Answer any FIVE full questions by choosing at least TWO from each part

PART-A

1. a. Distinguish between enhancement and depletion mode MOSFETs. Discuss nMOS enhancement mode transistor operation in different regions and also derive the equation for current I_{DS} in linear region. **09**

   b. With necessary circuit diagram and expressions, narrate the body effect of an nMOS transistor. **05**

2. a. Obtain the DC transfer characteristics of a CMOS inverter and mark all the regions. Show the status of PMOS and NMOS transistors and also derive the expression for Vout in region D. **09**

   b. Calculate the mid-point voltage $V_M(V_{TH})$ values on the CMOS DC characteristics for the following β ratios: **05**

   i. $\frac{\beta_n}{\beta_p} = 1$ (ii) $\frac{\beta_n}{\beta_p} = 10$

   ii. $\frac{\beta_n}{\beta_p} = 0.1$,Assume V_{DD}=5V, V_{thn}=1V, V_{thp}=-1V

3. a. Describe in detail step-by-step procedure of P-well CMOS fabrication process. **09**

   b. What is Lambda (λ) based design rule? List the (λ)-based design rules for a CMOS layers. **05**

4. a. Draw the Schematic, stick diagram and Layout of a CMOS 2-input NAND gate. Use standard color/monochrome codes. **09**

   b. Define Sheet Resistance. Calculate the ON resistance of a CMOS inverter with $Z_{PU}=\frac{8\lambda}{2\lambda}$, $Z_{PD}=\frac{2\lambda}{2\lambda}$, R_{SN}=10KΩ and R_{SP}=2.5KΩ for a 5μm Technology. **05**

PART-B

5 a. Define Scaling of MOS Circuits. Explain different types of Scaling Models. **04**

   b. Find the scaling factors for the following: **10**

   i. Gate capacitance

   ii. Maximum Operating frequency

   iii. On Resistance

   iv. Saturation Current

   v. Carrier charge density in channel

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6. a. Explain the following logic styles with suitable example and also state its advantages : **08**

   i. Dynamic CMOS Logic

   ii. Pseudo – nMOS Logic

   b. Realize 2-input XOR & XNOR gate using following logics: **06**

   i. Pass Transistor Logic

   ii. Transmission Gate Logic

7. a. Design and realize 4x4 Serial Multiplier using CMOS logic. **09**

   b. Design and implement 4-bit Barrel Shifter using 4x4 Crossbar switch logic. **05**

8. a. Compare between SRAM and DRAM memories. Draw the circuit diagram of 6T SRAM Cell, explain the working principle of read and write operations. **09**

   b. Design and realize 4X4 NOR based ROM memory cells. **05**